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Video Generation Models are General-Purpose Vision Learners

A Pre-Trained Text-to-Video Diffusion Backbone Achieves State-of-the-Art Across Depth, Pose, Segmentation, and 3D Keypoint Tasks via GenCepion

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The shift from task-specific NLP models to powerful generalists began when researchers recognized that next-token prediction on large corpora incidentally learns representations useful for every downstream task. This paper argues that large-scale text-to-video generation plays an analogous role for computer vision: the spatiotemporal priors, vision-language alignment, and scalability of video generation pre-training transfer to a wide range of perception tasks with no task-specific architectural changes. GenCepion, the resulting framework, achieves state-of-the-art performance on depth estimation, surface normal prediction, camera pose estimation, expression-referring segmentation, and 3D keypoint detection—from a single frozen backbone.

AT A GLANCE

Problem: Computer vision still relies on task-specific architectures; a single backbone that matches specialized models across diverse tasks has not existed.

Why it matters: A single foundation model for vision reduces engineering overhead and may exhibit better cross-task generalization than ensembles of specialists.

Method: Freeze a pre-trained 7B text-to-video diffusion transformer; adapt to perception tasks via lightweight task-head fine-tuning with text-instruction task routing.

Result: State-of-the-art or near-SOTA on 5 of 7 diverse vision benchmarks; 38% improvement on camera pose over image-only diffusion baselines.

Evidence: ECCV 2026 accepted; evaluated on NYUv2, Taskonomy, CO3D, EmotionNet, Human3.6M, and two more.

The Big Picture

Computer vision tasks—depth estimation, surface normal prediction, camera pose estimation, semantic segmentation, human pose estimation—have traditionally been addressed by specialist architectures trained on task-specific datasets. Even with the advent of large vision-language models (VLMs), perception tasks requiring geometric 3D understanding remain largely the domain of specialized networks.

The analogy with NLP is instructive. Before GPT, the standard recipe was: (1) design an architecture for the task, (2) train on labeled data, (3) deploy. Next-token prediction on large corpora replaced step (1) and dramatically reduced the labeled data requirement for step (2). This paper asks: what is the vision analogue of next-token prediction?

The authors argue it is text-to-video generation. Generating realistic video requires the model to understand depth, 3D geometry, object motion, and scene layout—exactly the representations needed for downstream vision tasks. Large-scale video generation pre-training on hundreds of millions of clips thus incidentally instills the priors that specialist architectures must acquire from task-specific labels.

Why Existing Approaches Fall Short

Image diffusion models (Stable Diffusion, DALL-E 3) have been adapted for perception tasks by treating the output domain (depth maps, surface normals) as structured pseudo-images. But image-only pre-training lacks temporal information, producing representations that are weaker on 3D-geometric tasks: camera pose estimation (which requires cross-view geometric consistency) and 3D keypoint detection (which requires depth understanding) degrade by 38% and 12% respectively when using an image backbone instead of a video backbone.

Specialist models (Depth Anything v2, ViTPose) achieve strong results on their respective tasks but require separate architectures, training pipelines, and deployment instances. GenCepion replaces them with a single checkpoint—though specialists retain a slight advantage on tasks where large task-specific datasets are available.

Core Mechanism

GenCepion treats every perception task as a conditional generation problem. The input is an image (or video frame); the output is a structured target (depth map, surface normal map, pose heatmap, etc.) encoded as a pseudo-image in the same pixel space.

A text instruction (e.g., “produce a metric depth map of this scene”) routes the backbone to the appropriate output distribution. The diffusion backbone generates

the target pseudo-image in the usual denoising loop. A lightweight 3-layer MLP head applied to the final denoised latent maps it to the task-specific output format (metric depth values, unit normal vectors, etc.).

The key insight is that the backbone, pre-trained to generate any video frame conditioned on text, can generate any structured pseudo-image conditioned on a perception-task instruction—without any change to its weights.

Technical Formulation

Let I be the input image, τ the text task instruction, and $\hat{Y}$ the target structured output (a depth map, pose heatmap, etc.). GenCepion models:

$$p(\hat{Y} \mid I, \tau) = \int p(\hat{Y} \mid z_T, \tau) \prod_{t=1}^T p(z_{t-1} \mid z_t, I, \tau) dz_{1:T}$$

via DDIM sampling with $T = 50$ steps, where the backbone’s attention cross-attends to both the image I (via image conditioning) and the instruction τ (via CLIP text encoding). The MLP task head g_ϕ then maps the final latent z_0 to the task output: $Y = g_\phi(z_0)$.

Only ϕ (the MLP head, approximately 2M parameters) is fine-tuned on task-specific data; the backbone’s 7B parameters remain frozen.

Learning or Inference Procedure

1. Pre-train the text-to-video diffusion backbone on 500M video-text pairs at resolutions up to 1080p (not part of this paper; uses an existing backbone).
2. For each perception task, collect task-specific labeled data (e.g., NYUv2 for depth, Human3.6M for 3D keypoints).
3. Fine-tune the MLP task head for 10K steps with learning rate 1e-4; backbone frozen throughout.
4. At inference, run DDIM with 50 steps to denoise the target pseudo-image; apply the task head to the final latent.
5. Optionally, condition on a sequence of frames for tasks requiring temporal consistency (e.g., video depth estimation).

What the Guarantee Says

The paper does not provide formal guarantees, but the empirical evidence supports two strong claims: (1) video pre-training provides richer priors than image pre-training for 3D-geometric tasks (camera pose: 38% improvement; 3D keypoints: 12%); and (2) task-instruction text conditioning is a necessary component—removing it degrades performance by 8.3% on average, confirming it serves as a task router rather than mere noise.

Experimental Findings

Task (dataset)	GenCep.	Prior SOTA	Δ
Depth (NYUv2)	0.081	0.089	+9.0%
Normals (Taskonomy)	11.2°	12.8°	+12.5%
Camera pose (CO3D)	2.1°	3.4°	+38.2%
Expr. seg. (EmotionNet)	71.3 mAP	68.1	+4.7%
3D keypts. (H3.6M)	38.2 mm	41.3 mm	+7.5%

GenCepion is competitive or superior in all five categories. The largest gains appear on 3D-geometric tasks (pose, depth), consistent with the hypothesis that video pre-training instills strong geometric spatial priors absent from image-only baselines.

Ablations and Interpretation

Video vs. image backbone: Replacing the video backbone with Stable Diffusion 3.5 (image-only, comparable size) degrades camera pose estimation by 38% and 3D keypoints by 12%, confirming temporal pre-training as the source of 3D geometric priors.

Pre-training data scale: Halving the pre-training video volume degrades downstream AUC by 6.2% on average; doubling improves by 4.1%, confirming a log-linear scaling law between video volume and downstream perception quality.

Frozen vs. fine-tuned backbone: Full backbone fine-tuning improves depth estimation by 1.8% but degrades surface normals by 1.2%, suggesting dataset-specific overfitting; the frozen-backbone default balances cross-task generalization.

Text conditioning ablation: Removing the task-instruction prefix and using image-only conditioning degrades performance by 8.3% on average—the largest single ablation—confirming text as the task-routing mechanism rather than an incidental input.

Reference

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